8.2.5 Preconditioning

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Week 7-8 due Mar 10, 2025 16:38 IST

Video

matter of fact, it's a

matter of replacing your search direction by M inverse times the

residual. And you have the same net effect, where in the end, you end up with

the vector x that you want. However, there remains a problem. And that is this

matrix may not be symmetric positive-definite. And all of the

discussion we've had so far is for symmetric positive definite matrices. So

then either we need to generalize our

7:42 / 7:42

2.0x

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Reading assignment

1/1 point (graded)

Read Unit 8.2.5 of the notes. [LINK]

- [x] done

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Ill conditioning: oscillations drawn by Professor — 4

At around 0:46, Robert explains the problems of ill conditioning and how the...

Homework 8.2.5.1

1/1 point (graded)

Show that the following algorithms

$\textbf{Given :}\ A, b, x^{(0)},$	$\textbf{Given :}\ A, b, x^{(0)}, M$
$\quad M = LL^T$	
$\tilde{A} = L^{-1} A L^{-T}$	
$\tilde{b} = L^{-1} b$	
$\tilde{x}^{(0)} = L^T x^{(0)}$	
$\tilde{r}^{(0)} := \tilde{b} - \tilde{A}\tilde{x}^{(0)}$	$r^{(0)} := b - Ax^{(0)}$
$k := 0$	$k := 0$
$\textbf{while}\ r^{(k)} \neq 0$	$\textbf{while}\ r^{(k)} \neq 0$
$\quad \tilde{p}^{(k)} := \tilde{r}^{(k)}$	$\quad p^{(k)} := M^{-1} r^{(k)}$
$\quad \tilde{q}^{(k)} := \tilde{A}\tilde{p}^{(k)}$	$\quad q^{(k)} := Ap^{(k)}$
$\quad \tilde{\alpha}_k := \frac{\tilde{p}^{(k)\,T} \tilde{r}^{(k)}}{\tilde{p}^{(k)\,T} \tilde{q}^{(k)}}$	$\quad \alpha_k := \frac{p^{(k)\,T} r^{(k)}}{p^{(k)\,T} q^{(k)}}$
$\quad \tilde{x}^{(k+1)} := \tilde{x}^{(k+1)} + \tilde{\alpha}_k \tilde{p}^{(k)}$	$\quad x^{(k+1)} := x^{(k+1)} + \alpha_k p^{(k)}$
$\quad \tilde{r}^{(k+1)} := \tilde{r}^{(k)} - \tilde{\alpha}_k \tilde{q}^{(k)}$	$\quad r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)}$
$\quad x^{(k+1)} = L^{-T} \tilde{x}^{(k+1)}$	
$\quad k := k + 1$	$\quad k := k + 1$
$\textbf{endwhile}$	$\textbf{endwhile}$

compute the same values for $x^{(k)}$.

- [x] done

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